

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

A Project Report

on

GeoVision: Intelligent CCTV & Entry Control System

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“Grand Challenge Studio”

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ABSTRACT

Traditional entry control systems in educational and corporate environments often rely on manual verification or physical ID scanners, resulting in significant bottlenecks and operational delays during peak hours. To address this, GeoVision is proposed as an intelligent, automated entry control system that leverages existing CCTV infrastructure to provide real-time facial recognition and comprehensive crowd monitoring.

The system is designed for campus administrators, security personnel, and facility managers who require faster and more reliable access control. Built upon a robust technical stack including Python and the Django framework, GeoVision integrates OpenCV for sophisticated video processing and utilizes a vector database for high-precision facial feature matching.

Key outcomes of this deployment include the significant reduction of physical entry bottlenecks, the implementation of automated, seamless user registration workflows, and the provision of actionable, live crowd density metrics. Looking forward, the project roadmap includes the integration of liveness detection to prevent unauthorized spoofing, potential deployment on edge-computing platforms for increased scalability, and the incorporation of predictive behavioral analytics to enhance overall facility security.

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CHAPTER 1 - PROBLEM STATEMENT

1.1 Background

Physical security systems are increasingly shifting toward intelligent automation. Modern institutions require solutions that combine surveillance and access control in a way that improves safety while maintaining user convenience.

1.2 Existing Challenges

Current entry systems face several operational challenges:

- Manual ID verification consumes time and increases dependency on human effort.
- Fingerprint scanners and physical checkpoints often create queues during peak hours.
- Conventional CCTV systems are mostly reactive and used for post-incident review rather than proactive access control.

1.3 Problem Statement

The problem addressed in this project is the lack of a unified, proactive system that can use existing CCTV infrastructure for both secure identity verification and crowd monitoring. GeoVision aims to solve this by developing a real-time face-scanning and entry control system that automates secure access to facilities.

1.4 Scope

The project includes design of a facial registration workflow, development of a facial recognition pipeline for live video feeds, and creation of a monitoring dashboard for crowd density and recognized-user logs.

CHAPTER 2 - PROBLEM VALIDATION

2.1 Target Customers

Educational institutions, corporate offices, event organizers.

2.2 Validation Methodology

- Interviews with campus security staff.
- Observations of entry bottlenecks during peak hours.

2.3 Customer Feedback

Manual verification consumes 30% of operational time.

2.4 Validation Results

Validation Question	Positive	Negative
Is current entry verification too slow?	85%	15%
Would automated face-scanning improve efficiency?	92%	8%

2.5 Key Insights

Need for speed, accuracy, and integration with existing surveillance.

CHAPTER 3 - LITERATURE SURVEY

3.1 Existing Solutions

Most currently available solutions focus on only one side of the problem. Video Management Systems (VMS) primarily handle surveillance, while RFID readers or biometric scanners are usually dedicated to access control. Few solutions combine both functions effectively in a lightweight and web-integrated manner.

3.2 Research Papers Reviewed

Existing solutions often suffer from high infrastructure cost for deployment, limited web-based administrative control, and poor integration between identity verification and crowd analytics.

3.3 Industry Solutions

A more practical solution should combine surveillance intelligence, real-time authentication, and lightweight deployment in one platform. This is especially important for campuses and institutions that already possess CCTV infrastructure.

3.4 Research Gap

Lack of systems handling both crowd density and individual identity in a unified workflow.

CHAPTER 4 - PRODUCT REQUIREMENTS

4.1 Stakeholders

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The main stakeholders are system administrators, registered end-users (students or employees), and security personnel.

4.2 Functional Requirements

ID	Requirement	Priority
FR1	The system must capture and register facial encodings through a web interface.	High
FR2	Real-time CCTV feeds must authenticate registered users instantly.	High
FR3	The dashboard must display current crowd density metrics.	Medium

4.3 Non-Functional Requirements

High accuracy, low latency, usability, compatibility.

4.4 User Stories

Example:

As a security officer, the system should provide a real-time log of recognized individuals so that occupancy can be monitored effectively.

As a user, the registration process should be quick and simple so that the face can function as a secure entry pass.

4.5 Success Metrics

- Recognition accuracy >95%
- 80% reduction in wait time

CHAPTER 5 - SOLUTION DESIGN

5.1 Proposed Solution Overview

GeoVision acts as middleware between camera hardware and institutional databases. It processes video frames, extracts facial features, verifies identities, logs recognized users, and generates occupancy-related insights.

5.2 System Architecture

Camera -> Image Processing -> Embedding Extraction -> Vector Search -> Database/Dashboard

5.3 Technology Stack

Layer	Technology
Backend	Python, Django
Vision Core	OpenCV, Dlib / face_recognition
Database	Vector Database / SQLite

5.4 AI Components

The AI core of GeoVision relies on pre-trained convolutional neural network models for facial landmark detection and feature extraction.

- Facial registration module.
- Similarity search in vector database.

5.5 Database Design

Users Table (ID, Name, Face_Encoding_Vector), Access_Logs (EntryID, UserID, Timestamp).

5.7 Security Considerations

CHAPTER 6 - IMPLEMENTATION

6.1 Development Environment

The project was developed using Python 3.x with isolated virtual environments. Git was used for source control.

6.2 Modules

- Registration Module, Vision Module, Analytics Module, Admin Dashboard Module.

6.3 Screenshots

Include application screenshots.

6.4 Demo Walkthrough

1. Arrival at geofence. 2. Biometric verification activation. 3. Local embedding extraction. 4. Vector similarity check. 5. Entry logging.

6.5 Challenges Encountered

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Technical hurdles included optimizing model latency and handling varied ambient lighting.

CHAPTER 7 - RESULTS

7.1 Experimental Setup

Tested under different lighting conditions and with multiple simultaneous faces.

7.2 Evaluation Metrics

7.3 Results

Latency: < 500ms; Positive Rate: 96.5%; Simultaneous Detections: 15 faces.

7.4 User Feedback: Positive reception for speed and ease of use.

CHAPTER 8 - PROMOTION VIDEO

8.1 Target Audience

8.2 Video Link

[YouTube/Drive Link]

8.3 Marketing Message

Smart entry, zero friction: Automate your campus security with GeoVision.

CHAPTER 9 - CONCLUSION & FUTURE PLANS

9.1 Summary

The project successfully demonstrated a unified face-scanning and crowd monitoring workflow.

Summarize achievements.

9.2 Outcomes

Discuss project impact.

9.3 Limitations

Current constraints.

9.4 Future Enhancements

Features

Scaling and AI Strategy

Roadmap

REFERENCES

Example:

[1] Author Name, "Paper Title," Journal Name, Year.

[2] Website Name, URL, Accessed Date.

APPENDICES

Appendix A – Customer Survey Questionnaire

Appendix B – Screenshots

Appendix C – Source Code Repository

Appendix D – User Manual

Appendix E – Promotion Video QR Code

